CS411 Lab II

GradMap vs DegreeWorks

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1 Introduction

1.1 Purpose

The purpose of this document is to be a guideline for the development of GradMap. It will serve as a blueprint outlining the basic functions of the program as a whole, as well as how to implement each user role and the basic systems on which to build those functions. This is the developer bible for GradMap.

1.2 Scope

GradMap is a program in development by a small team that is a mission to provide a solution to one of the most common issues encountered by university students: managing a classload that is not only able to fit one's needs, but ensures that any user will be able to track the pace at which they are graduating, schedule for upcoming semesters, and ensure that all students using the program are able to graduate in an efficient and effective manner. GradMap intends to provide an easy to read flowchart that lists all prerequisites and corequisites in an accessible and clear way.

Running Head: GradMap

1.3 Definitions, Acronyms, Abbreviations

GradMap

An interactive application designed to assist students and academic advisors in visualizing academic progress, managing course completion, and tracking degree requirements in real-time.

Data Management

The process of collecting, storing, and updating student information and progress data efficiently and securely within the GradMap system.

Real-Time Alerts

Notifications sent instantly to users when significant events occur, such as missing prerequisites, approaching graduation requirements, or changes in course availability.

Real-Time Data

The immediate access and updating of academic information within the GradMap dashboard, ensuring that users always view the most recent data.

Sensor Threshold

A system-defined limit used in GradMap's backend monitoring tools that triggers alerts when performance metrics or usage patterns exceed expected values.

User Dashboard

The central interface of GradMap that allows users to view degree progress, upcoming courses, alerts, and academic milestones all in one location.

Running Head: GradMap

Data Visualization

The use of interactive graphs and charts in GradMap to represent academic progress, credit completion, and GPA trends in a visual and understandable way.

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1.5 Overview

The next section of this document will go over the basic framework of the implementation of GradMap and how the different mechanisms will interact with one another, as well as the basic user roles and responsibilities allotted to said roles.

2 Overall Description

Figure 1: GradMap WIP Logo

GradMap is a web-based application intended for use by students to assist in developing and maintaining a proper schedule per semester and keeping track of their progress through their respective degree's curriculum. The system intends to address a multitude of issues that normally arise when students are managing their future courses, including but not limited to: Unfulfilled Prerequisites, life and school overlap, class exclusivity, and so much more. The main objective of this program is to increase the accessibility of streamlining this arduous process for university students.

Running Head: GradMap

2.1 Product Perspective

GradMap will use course and student data provided by the university to assist students in building a proper schedule that fits their individual needs and constraints. This system is built to be integrated into current university systems, rather than to replace a facet that is currently in use. The program has a user interface that the user interacts with and performs their functions, the applications beneath that will process the user interface and update the data layer with current information provided by the user.

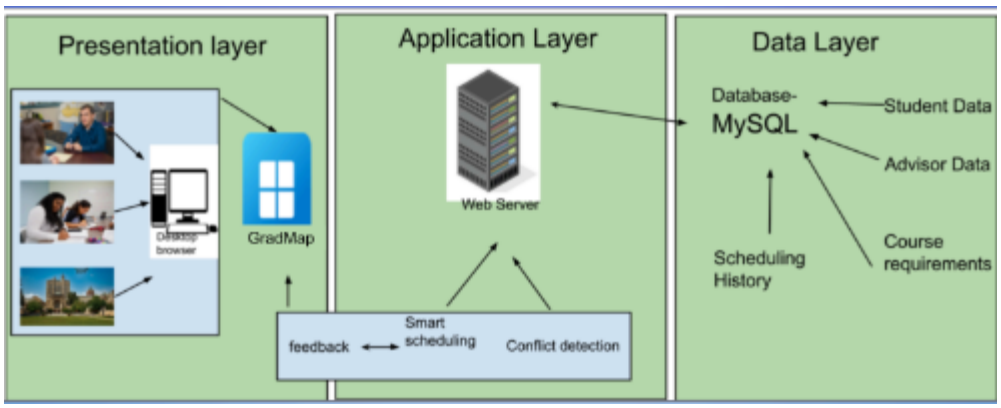

Figure 2: GradMap MFCD

2.2 Product Functions

The GradMap prototype will display dummy information given by an example university to show how the program would function by using an actual university database. This can be configured further to be applicable to different universities different databases. GradMap will generate possible student schedules for upcoming semesters to streamline the process for students, additionally GradMap will allow students and advisors to connect much more easily than they currently are able to.

2.3 User Characteristics

The student accounts will be created by the university for students and will be linked to student emails, the interactivity will be streamlined similarly to most websites, where it can be navigated through buttons and searching.

The advisor accounts will mostly have access to their list of students, and will be able to draft emails for their students to read, as well as being able to choose whether the message or email is personalized or to be sent to all of their students.

The administrator accounts will be the only accounts that are able to directly edit the database structure, changing course information, advisor assignments, and degree requirements.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

N/A